ISA/IEC 62443-3-3 / 4-2 • SL-3 TESTING CHEAT SHEET

Rockwell/Allen-Bradley 1734-AENTR (POINT I/O) & 1738-AENTR (ArmorPOINT I/O) EtherNet/IP Adapter · Component type: **Embedded Device (EDR) - EtherNet/IP remote-I/O adapter + Modbus/EtherNet-IP conduit gateway** · Target SL = SL 3

Where to work	Configure in **Studio 5000 Logix Designer** (the adapter + its I/O appear under the controller I/O tree via the **Add-On Profile** - Module Properties: **Electronic Keying**, connection/RPI); IP is set with the **BOOTP-DHCP Tool** / rotary switches / **FactoryTalk Linx** (RSLinx); the adapter has an **embedded web server** (HTTP diagnostics/config) that must be **disabled after commissioning** (CISA ICSA-21-063-01). Access control is enforced at the **Logix controller + FactoryTalk Security**, not on the adapter. The **Modbus↔EtherNet/IP conversion** is done by a companion gateway - **ProSoft PLX31/PLX51** (configured in **ProSoft Configuration Builder, PCB**) or in-chassis **1756-MVI56E-MCM** - assess it in the gateway rows.
Ports / paths	**Adapter (EtherNet/IP):** CIP explicit **TCP 44818**, implicit I/O **UDP 2222**, ListIdentity **UDP 44818**; embedded web **HTTP 80** (**no HTTPS, no CIP Security** on 1734/1738-AENTR); **BOOTP/DHCP UDP 67/68**; **SNMP UDP 161** (if enabled). **Modbus↔ENIP gateway (ProSoft PLX3x): Modbus TCP 502**, EtherNet/IP **44818 / UDP 2222**, PCB config **UDP 50000** (discovery) + **UDP 3000** (config/diag), web **HTTP 80**.
Firmware / CVEs	1734-AENTR (POINT I/O) / 1738-AENTR (ArmorPOINT I/O) dual-port adapters. **Patch the web-interface auth-bypass**: **CISA ICSA-21-063-01** - the 1734-AENTR (Series B & C) web interface mishandles authentication for HTTP POST, letting an unauthenticated remote attacker modify configuration; fix = updated firmware **and** disable the web server. No CIP Security on this adapter (migration path: 1756-EN4TR / CIP-Security-capable devices). Track the gateway (ProSoft / 1756-MVI) against ProSoft & CISA advisories.
How to read	Default read-only (Studio 5000 read-only project review; read-only CIP/EtherNet-IP + Modbus enumeration). Each row = native step-by-step + command-level external-tool validation + pass. Write/exploit tools = isolated env + written authorization. This is a legacy I/O adapter: many rows are honestly Compensating / N-A-at-device.

FR 1 — Identification & Authentication Control (IAC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 1.1	Human user identification & auth	The adapter has **no native user accounts**. Human access is authenticated at the **Logix controller + FactoryTalk Security** (who may go online / download the I/O config) - confirm that is enforced. The adapter's **embedded web server** had a broken-auth flaw (ICSA-21-063-01), so **disable the web server** after commissioning and patch firmware. **Test:** browse `http://` -> web disabled/patched (no unauthenticated config POST); going online to change the I/O config requires FactoryTalk/controller credentials. **Ext. tools:** FactoryTalk Security (policy review); `curl -i http:///` (expect closed); `nmap -p 80,44818 .`	Engineering authenticated at controller/FactoryTalk; adapter web server disabled + patched; no anonymous config.	EDR - no device auth; compensating.
SR/CR 1.2	Device & software-process I&A;	EtherNet/IP CIP on this adapter is **unauthenticated** (no CIP Security). Compensate: set **Electronic Keying = Exact Match** (or Compatible) in the Studio 5000 Module Properties so the controller binds the exact adapter identity/revision, and restrict CIP peers by segmentation/firewall. **Test:** from an unauthorized host, issue CIP ListIdentity / open a class-3 connection -> blocked by segmentation. **Ext. tools:** Studio 5000 (Electronic Keying); `nmap --script enip-info -p 44818` ; Wireshark `enip || cip`.	Exact-match keying binds device identity; unauthenticated CIP compensated by keying + segmentation.	No CIP Security - Compensating.
SR/CR 1.3	Account management	No accounts on the adapter; manage engineering identities in **FactoryTalk Security** (least-privilege, named users, no shared logins) and keep custody of the Studio 5000 project (which holds the I/O/adapter config). For the **Modbus↔ENIP gateway**, manage its web/PCB accounts and remove default credentials. **Ext. tools:** FactoryTalk Security (user/role export); gateway web/PCB account review; project-custody record.	Named least-privilege engineering accounts (FactoryTalk); gateway defaults removed; project protected.	
SR/CR 1.4	Identifier management	Confirm a unique device identity: unique **IP / node address** (rotary switches or BOOTP/DHCP reservation), unique CIP identity (product code/serial), and unique gateway identity; no reuse across cells. **Ext. tools:** Studio 5000 (module identity); `nmap --script enip-info -p 44818` ; network inventory.	Unique device/gateway identifiers; no reuse.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 1.5 / 1.7	Authenticator management & strength	The adapter stores **no device credentials**; authenticator strength is enforced at **FactoryTalk Security** (engineering) and at the **gateway** - change the ProSoft/MVI default web/config password to a strong one and manage its lifecycle. **Ext. tools:** FactoryTalk Security (password policy); gateway web (change default creds; policy review).	Strong FactoryTalk + gateway credentials; defaults changed; managed lifecycle.	Adapter: no credentials.
SR/CR 1.6	Wireless access management	**N/A.** Wired EtherNet/IP adapter (POINT I/O in-cabinet; ArmorPOINT on-machine) with no native wireless interface. **Ext. tools:** Confirm no wireless; upstream network governs.	N/A - justification recorded (wired device).	N/A
SR/CR 1.8 / 1.9	PKI certificates & key strength	**N/A / limited at device.** 1734/1738-AENTR have no device PKI/TLS (HTTP-only, no HTTPS, no CIP Security). Record N/A; provide confidentiality/integrity via segmentation and, where required, migrate the conduit to a CIP-Security-capable path (e.g. 1756-EN4TR) or terminate TLS at the SCADA layer. **Ext. tools:** Confirm no configurable device certificate; Wireshark (web = HTTP cleartext; CIP unauthenticated).	N/A/limited device PKI; compensating segmentation / TLS upstream.	N/A/Partial.
SR/CR 1.10	Authenticator feedback	**N/A at adapter** (no device login). Confirm FactoryTalk logon and the gateway web login mask the secret (no cleartext echo). **Ext. tools:** Visual verification (FactoryTalk / gateway web).	Authenticators obscured at FactoryTalk/gateway (N/A at adapter).	N/A at device.
SR/CR 1.11	Unsuccessful login attempts	The adapter has **no login / no lockout**. Enforce lockout at **FactoryTalk Security** (engineering) and the gateway web (if supported); the adapter's exposure is removed by disabling the web server + segmentation; monitor failed CIP/Modbus attempts. **Ext. tools:** FactoryTalk Security (lockout policy); gateway web (lockout); OT IDS on the cell segment.	Lockout at FactoryTalk/gateway; adapter web disabled; attempts monitored.	Adapter: no lockout - Compensating.
SR/CR 1.12	System use notification	No banner on the adapter/embedded web; enforce the approved use notice on the Studio 5000 EWS (GPO), the SCADA HMI, and the gateway web login where supported (compensating). **Ext. tools:** EWS `gpresult /h gp.html`; HMI / gateway banner review.	Use notice at EWS/HMI/gateway (compensating at adapter).	Compensating.
SR/CR 1.13	Access via untrusted networks	Segment the adapter into a cell zone; firewall **44818/2222/80/161**; no EtherNet/IP from untrusted networks; VPN for cross-zone engineering. The **Modbus↔ENIP gateway is a boundary device** - firewall the **Modbus 502** side separately from the EtherNet/IP cell. **Test:** from an untrusted VLAN, reach 44818 / 502 -> must fail. **Ext. tools:** `nmap -Pn -sTU -p 44818,2222,80,161,502` from the untrusted side (expect filtered); firewall rule review.	Adapter + gateway unreachable from untrusted networks; brokered via VPN + firewall.	

FR 2 — Use Control (UC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 2.1	Authorization enforcement	The adapter performs **no independent authorization** - it presents I/O to the owning controller. Confirm authorization is enforced at the **Logix controller + FactoryTalk Security** (roles: who can go online / download / edit I/O), and that **Electronic Keying** prevents an unauthorized/mismatched module. For the gateway, restrict which Modbus registers are writable via its map. **Test:** as a read-only FactoryTalk user, attempt to download / change the I/O config -> denied. **Ext. tools:** FactoryTalk Security (role review); Studio 5000 (keying); gateway (Modbus map write scope).	Least privilege enforced at controller/FactoryTalk; keying blocks mismatched modules.	Delegated to controller.
SR/CR 2.2	Wireless Use Control	**N/A.** No wireless interface on this device; wireless access is managed at the network layer.	N/A - justification recorded (not assessed as a control).	N/A
SR/CR 2.3	Use control for portable/mobile devices	Control the Studio 5000 laptop and removable media (OS device-control); only authorized projects/firmware loaded. **ArmorPOINT (1738)** is on-machine (IP67) and physically exposed - control physical access to its ports; POINT (1734) sits in a cabinet. **Ext. tools:** EWS removable-media GPO; physical inspection (ArmorPOINT on-machine ports).	Engineering media controlled; on-machine adapter physical access controlled.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 2.4 (+ EDR 2.4)	Mobile code	No general-purpose mobile code; confirm only Rockwell firmware runs on the adapter and the embedded web content is vendor read-only - **disable the web server** (it is the ICSA-21-063-01 exposure). The gateway runs only vendor firmware. **Ext. tools:** Studio 5000 / gateway (firmware review); confirm web server disabled.	Only trusted vendor firmware runs; web disabled; no untrusted mobile code.	EDR-specific.
SR/CR 2.5	Session lock	**N/A at adapter** (no interactive session). Enforce workstation lock on the Studio 5000 EWS (OS screen lock/GPO); gateway web idle timeout if supported. **Ext. tools:** EWS inactivity-lock GPO; gateway web timeout review.	EWS locks on inactivity (N/A at adapter).	Compensating.
SR/CR 2.6 / 2.7	Remote session termination & concurrency	Confirm CIP connections can be closed and the adapter **bounds concurrent CIP connections** (connection limit) so it cannot be exhausted; the gateway bounds concurrent Modbus/ENIP connections. **Ext. tools:** Studio 5000 / `nmap --script enip-info` (connection info); gateway connection review; Wireshark (teardown).	Connections bounded/terminable at adapter + gateway.	
SR/CR 2.8 / 2.12	Auditable events & non-repudiation	The adapter has **no native security/syslog log**. Capture configuration-change attribution at the **Logix controller + FactoryTalk AssetCentre** (audit/change logs), and log CIP/Modbus connection + write events with a **CIP/Modbus-aware OT IDS**; forward to the SIEM. **Ext. tools:** FactoryTalk AssetCentre (change/audit); OT IDS (CIP/Modbus) on a SPAN + SIEM.	Changes attributable at controller/AssetCentre; adapter/gateway events captured by IDS.	No native syslog - Compensating.
SR/CR 2.9 / 2.10	Audit storage capacity & failure handling	Because logging is off-device (AssetCentre / IDS / SIEM), confirm those are sized/retained and alert on collection failure so adapter/gateway events are not silently lost. **Ext. tools:** AssetCentre / SIEM retention + source-silent alert review.	Off-device logs sized/retained; audit-failure alerting defined.	Compensating.
SR/CR 2.11	Timestamps	The adapter has no user RTC; event times come from the **controller / SCADA**. Confirm the controller and IDS/SIEM are time-synced (NTP/CIP time) so adapter/gateway events correlate. **Ext. tools:** Studio 5000 / controller time (CIP); `nmap -sU -p 123` / Wireshark `ntp`.	Reliable synchronized time at controller/IDS; events correlate.	
EDR 2.13	Physical diagnostic & test interfaces	Control physical interfaces: **POINT I/O** adapter in a locked cabinet; **ArmorPOINT** on-machine - secure/seal its connectors; protect the **rotary address switches** (set + record node) and the Ethernet ports; disable the web server and any unused service (SNMP) so no diagnostic backdoor remains. **Ext. tools:** Physical inspection (cabinet/seals, rotary-switch cover); `nmap -sU -p 161` (SNMP off if unused).	Physical/diagnostic interfaces controlled; web + unused services disabled.	EDR-specific.

FR 3 — System Integrity (SI)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 3.1 / 4.1	Communication integrity & confidentiality	EtherNet/IP CIP + the embedded web are **unauthenticated cleartext** (no CIP Security/HTTPS on this adapter) - integrity/confidentiality come from **containment**. Keep CIP/I/O inside a segmented cell zone; the **Modbus↔ENIP gateway is the conduit** - place a firewall between the Modbus network and the EtherNet/IP cell so no cleartext control crosses a zone boundary uncontrolled. **Test:** confirm CIP (44818/2222) and Modbus (502) stay in-zone (no cleartext crossing the boundary uncontrolled). **Ext. tools:** Wireshark `enip || cip || modbus` (cleartext, in-zone only); firewall rule review.	Control comms contained/segmented; conduit firewalled; no uncontrolled cleartext across boundaries.	No CIP Security - Compensating.
EDR 3.2	Protection from malicious code (component)	Confirm only Rockwell firmware runs on the adapter (no code path beyond it). Patch to the fixed firmware for **ICSA-21-063-01**; obtain firmware from Rockwell and load via **ControlFLASH Plus** (newer Rockwell images are digitally signed and verified on load). The gateway runs only signed/verified ProSoft firmware. **Ext. tools:** ControlFLASH Plus (version + signature); Studio 5000 (module firmware); Rockwell PSA / CISA advisory review.	Only vendor firmware runs; ICSA-21-063-01-patched; provenance/signature verified.	EDR-specific.
SR/CR 3.3	Security functionality verification	After any change re-verify: web server disabled, unused services off (SNMP), Electronic Keying set (Exact/Compatible), firmware patched, segmentation/conduit-firewall intact, and the gateway config, against the hardening baseline; compare the Studio 5000 project. **Ext. tools:** Studio 5000 (settings vs baseline); `nmap -sTU -p 44818,2222,80,161,502` (only required reachable).	Security settings match the hardening baseline after change.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 3.4	Software & information integrity	The adapter/I/O configuration lives in the **Studio 5000 project** - protect it (host ACLs, version control); Electronic Keying + controller download-control prevent an unauthorized module/config change. Protect and hash the **gateway (PCB) config file** to detect drift. **Ext. tools:** Studio 5000 (online-vs-offline compare); `certutil -hashfile SHA256` for drift.	Config integrity protected; unauthorized change prevented/detected.	
SR/CR 3.5	Input validation	Confirm the adapter tolerates malformed **CIP / EtherNet/IP** frames and the gateway tolerates malformed **Modbus** frames without faulting or dropping I/O. **ISOLATED only:** fuzz on a bench unit and watch connection/I/O health. **Ext. tools:** CIP/EtherNet-IP + Modbus fuzzer (`boofuzz`) on a bench unit; Wireshark; watch adapter/gateway status LEDs + connection state.	No fault or loss of I/O on malformed input.	Do NOT fuzz an in-service unit.
SR/CR 3.6	Deterministic output	Verify configured **fault/fallback behavior** for adapter outputs on connection loss/timeout - outputs assume the configured safe state (fault action / hold-last) when the CIP connection times out; confirm RPI/connection timeout and DLR ring behavior. **Ext. tools:** Studio 5000 (output-module fault action / connection-timeout config); simulate CIP comm loss -> defined output state.	Outputs go to a defined/fail-safe state on connection loss.	
SR/CR 3.7	Error handling	Confirm graceful fault handling: adapter **module/network status LEDs** + controller connection diagnostics report faults without unsafe behavior or info leak. **Ext. tools:** Studio 5000 (module diagnostics); adapter LED/status review.	Graceful fault handling; no unsafe state or info leak.	
SR/CR 3.8	Session integrity	**N/A / limited.** No authenticated/encrypted session on the adapter (unauthenticated CIP, no CIP Security); integrity is provided by containment. Record N/A + compensating segmentation. **Ext. tools:** Wireshark (confirm in-zone containment; no replay path across boundary).	N/A/limited at device; compensating segmentation recorded.	N/A/Partial.
SR/CR 3.9	Protection of audit information	Since audit is off-device, protect the AssetCentre / IDS / SIEM store and restrict who can read/clear it. **Ext. tools:** AssetCentre / SIEM access review.	Audit records protected from unauthorized read/modify/delete.	
EDR 3.10	Support for updates	Update adapter firmware via **ControlFLASH Plus** with firmware from Rockwell; verify signature/version; plan in a maintenance window with rollback to the prior revision; apply the ICSA-21-063-01 fix. Update the gateway firmware from ProSoft likewise. **Ext. tools:** ControlFLASH Plus (update flow + signature); Rockwell PCDC provenance; ProSoft firmware for the gateway.	Controlled signed updates from a verified source, CVE-patched, with rollback.	EDR-specific.
EDR 3.11	Physical tamper resistance/detection	Keep **POINT I/O** in a locked/sealed cabinet; secure the **ArmorPOINT** on-machine unit and seal unused connectors; control the rotary switches and Ethernet ports; config changes are visible in Studio 5000 / AssetCentre. **Ext. tools:** Physical inspection (cabinet/seal, ArmorPOINT connectors, rotary cover); AssetCentre change review.	Cabinet/on-machine unit secured; interfaces controlled; changes logged.	EDR-specific.
EDR 3.12	Provisioning product-supplier roots of trust	Newer Rockwell firmware is **digitally signed** and verified by ControlFLASH Plus (product-supplier root of trust); confirm signed firmware / provenance for the loaded revision. Older AENTR revisions may predate signed images - record and compensate with provenance discipline (3.10), patching, and segmentation. **Ext. tools:** ControlFLASH Plus (signature verify); Rockwell PCDC provenance.	Signed-firmware supplier RoT where supported; provenance verified; gaps recorded.	EDR-specific/Partial.
EDR 3.13	Provisioning asset-owner roots of trust	**Partial/N-A at device.** 1734/1738-AENTR provide no asset-owner certificate provisioning (no CIP Security/PKI). Record the gap; asset-owner trust is enforced at the network layer (VPN/firewall/segmentation); migrate the conduit to a CIP-Security-capable device (1756-EN4TR) where asset-owner RoT is required. **Ext. tools:** Confirm no device certificate/CIP-Security provisioning; segmentation/migration review.	No asset-owner RoT at device; compensating controls / migration path recorded.	Partial/N-A.
EDR 3.14	Integrity of the boot process	Signed-firmware load via ControlFLASH Plus provides load-time integrity on supported revisions; there is **no published secure-boot attestation** for the adapter. Record the gap; mitigate via controlled firmware provenance (3.10), patching, physical control, and monitoring. **Ext. tools:** ControlFLASH Plus (firmware signature/version); confirm no secure-boot attestation.	Load-time firmware integrity where supported; no attestation - compensating controls recorded.	EDR-specific/Partial.

FR 4 — Data Confidentiality (DC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 4.2	Information persistence	On decommission/repurpose, **reset the adapter to out-of-box** (clear IP/config) and clear the I/O config from the Studio 5000 project; for the gateway, clear/erase its config (PCB) so Modbus maps and network config are not recoverable; sanitize any removable media. **Ext. tools:** Adapter reset-to-defaults; Studio 5000 (remove module); gateway config erase.	Adapter/gateway config purged on decommission or reuse.	
SR/CR 4.3	Use of cryptography	**N/A / limited.** No comms crypto on the adapter (no HTTPS/CIP Security). Rely on segmentation and TLS at the SCADA/EWS layer; **disable the embedded web**. Where confidentiality is required on the conduit, migrate to a CIP-Security-capable path. **Ext. tools:** Confirm no device crypto; confirm web disabled; Wireshark (no cleartext across boundary).	No device crypto - compensating TLS/segmentation upstream; web disabled.	N/A/Partial.

FR 5 — Restricted Data Flow (RDF)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 5.1 / 5.2	Network segmentation & zone boundary	Primary control. Confirm the adapter sits in a segmented cell zone; only required flows cross (CIP 44818/2222 to the owning controller). The **Modbus↔ENIP gateway is the conduit between two protocol networks** - enforce a firewall at that boundary and put the gateway's two interfaces on **different subnets**; only the intended Modbus 502 / EtherNet/IP flows cross. **Ext. tools:** `nmap -sTU -p 44818,2222,80,161,502` per zone; firewall/conduit rule review.	Adapter in a segmented zone; conduit gateway firewalled with dual-subnet separation; only required flows cross.	
SR/CR 5.3	General-purpose person-to-person comms restrictions	**N/A.** Single-purpose I/O adapter / protocol gateway with no general-purpose person-to-person messaging. **Ext. tools:** Confirm no such services.	N/A - justification recorded.	N/A
SR/CR 5.4	Application partitioning	For the **gateway**, its core job is partitioning two protocol networks - confirm the Modbus side and the EtherNet/IP side are separated (dual interfaces on different subnets + firewall) and only mapped registers cross. For the adapter alone, single-purpose - N/A. **Ext. tools:** Gateway interface/subnet + Modbus-map review; `nmap` per interface.	Gateway partitions the two protocol networks; adapter single-purpose (N/A).	Gateway-specific.

FR 6 — Timely Response to Events (TRE)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 6.1 / 6.2	Audit log accessibility & continuous monitoring	The adapter has no native syslog, so monitoring is off-device: confirm a passive **CIP/Modbus-aware OT IDS** (Nozomi/Dragos) on the cell segment + the conduit captures connection/write/CVE-exploitation events and forwards to the SIEM; **FactoryTalk AssetCentre** holds config-change/audit records that are exportable. **Ext. tools:** Passive OT IDS (CIP/Modbus-aware) on a SPAN; SIEM; AssetCentre change export.	Adapter/gateway events monitored off-device in near-real-time; changes exportable.	No native syslog - Compensating.

FR 7 — Resource Availability (RA)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 7.1	Denial-of-service protection	Confirm DoS resilience: the adapter bounds CIP connections but is sensitive to EtherNet/IP floods / CIP connection exhaustion - use cell-firewall rate controls and **DLR** for media resiliency; the gateway bounds Modbus connections. **Ext. tools:** ISOLATED bench: `hping3 --flood -p 44818` / CIP connection storm; watch adapter connection/I/O state + DLR.	Resources bounded; I/O continues under load; ring resiliency confirmed.	
SR/CR 7.2	Resource management	Confirm bounded resources: max CIP connections + appropriate **RPI** sizing on the adapter, and Modbus connection limits on the gateway, so a single service cannot starve I/O. **Ext. tools:** Studio 5000 (connection/RPI); `nmap --script enip-info`; gateway connection limits.	Resources bounded; no exhaustion-based loss of I/O.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 7.3 / 7.4	Backup & recovery	Confirm the **Studio 5000 project** (holds adapter/I/O config) and the **gateway PCB config** are backed up (versioned, offline) and a restore is tested on a spare adapter/gateway for a known-good rebuild. **Ext. tools:** Backup repository; `certutil -hashfile SHA256`; restore test (spare unit).	Backups current and recovery validated.	
SR/CR 7.5	Emergency Power	**N/A.** Emergency/backup power is provided at the facility/UPS level, not the device; confirm the device retains its security state across a power transfer.	N/A - justification recorded (not assessed as a control).	N/A
SR/CR 7.6 / 7.7	Network & security settings + least functionality	Harden: **disable the embedded web server**, disable unused services (SNMP), set Electronic Keying, patch firmware (ICSA-21-063-01); on the gateway, **disable its web server after configuration**, change default credentials, and restrict to the required Modbus/ENIP flows. **Ext. tools:** Adapter/gateway web (services off); Studio 5000 (keying); `nmap -sTU -p 80,161,44818,2222,502` (only required reachable).	Least functionality; web + unused services disabled; keying set; patched.	
SR/CR 7.8	Component inventory	Inventory the adapter (1734-AENTR / 1738-AENTR catalog + series + firmware) and the gateway (ProSoft PLX / 1756-MVI model + firmware) so vulnerabilities track against **Rockwell PSA / CISA** (ICSA-21-063-01) and **ProSoft** advisories. **Ext. tools:** `nmap --script enip-info -p 44818` ; Studio 5000 inventory; Rockwell PSA / ProSoft advisory feeds.	Accurate adapter/gateway/firmware inventory maintained; advisories tracked.	

External Test Tools — how to use

Tool / method	Verifies	How to use — steps / example	Constraints
Studio 5000 Logix Designer (AOP)	Native config & keying review (all FRs)	Open the project read-only; the adapter + I/O appear under the controller I/O tree (Add-On Profile) - review Module Properties (Electronic Keying, connection/RPI), module diagnostics, firmware revision. Export as the baseline.	Read/authorized project copy.
FactoryTalk Security / AssetCentre	Engineering I&A, authorization & audit (1.1, 1.3, 2.1, 2.8, 6.x)	Where human access is actually enforced (roles: go-online/download); AssetCentre holds config-change/audit records - export them.	Read-only review.
BOOTP-DHCP Tool / FactoryTalk Linx (RSLinx)	IP assignment & browse (1.2, 1.4)	Set/reserve the adapter IP; browse the EtherNet/IP tree; confirm device identity. Prefer a DHCP reservation or disable BOOTP after commissioning.	Authorized; commissioning.
ControlFLASH Plus	Firmware update, signature & provenance (3.2, 3.10, 3.12)	Update the adapter to the ICSA-21-063-01-fixed firmware from Rockwell (PCDC); confirm signature/version; keep the prior revision for rollback.	Maintenance window.
Embedded web server / curl	Web-exposure check (1.1, 2.4, 7.6)	`curl -i http:///`; confirm the web server is DISABLED (ICSA-21-063-01) after commissioning; no unauthenticated config POST.	Read-only; authorized.
ProSoft Configuration Builder (PCB) / gateway	Modbus↔ENIP conduit config (1.3, 3.4, 5.x)	Configure/inspect the ProSoft PLX (or 1756-MVI) Modbus↔EtherNet/IP map; change default web password; disable the web server after config; dual-subnet separation. Discovery UDP 50000 / config UDP 3000.	Authorized; back up the config.
nmap	Port/service exposure, EtherNet/IP ID & segmentation (1.x, 3.1, 5.x, 7.x)	`nmap -sTU -p 44818,2222,80,161,502` ; `nmap --script enip-info -p 44818` ; scan per zone.	Read-only; authorized.
Wireshark / tcpdump	Comms integrity & identity (1.2, 3.1, 3.8)	Filter `enip || cip || modbus`; confirm cleartext CIP/Modbus stay in-zone; inspect the conduit boundary.	Passive; SPAN/tap only.
CIP/Modbus fuzzer / hping3 (isolated)	Input validation & DoS resilience (3.5, 7.1)	`boofuzz` CIP/Modbus on a bench unit; `hping3 --flood -p 44818` storm; watch connection/I/O + DLR state.	ISOLATED bench; written authorization.
Passive OT IDS (CIP/Modbus-aware)	Monitoring (2.8, 6.x) - replaces native syslog	Nozomi/Dragos on a SPAN to capture CIP/Modbus connection/write + CVE-exploitation attempts; forward to the SIEM.	Passive; coordinate with SOC.

Tool / method	Verifies	How to use — steps / example	Constraints
Rockwell PSA / CISA / ProSoft advisories	Vuln tracking (3.2, 3.10, 7.8)	Track adapter advisories (CISA ICSA-21-063-01) and gateway (ProSoft / 1756-MVI) advisories + required firmware.	Reference.

Controlled - Project Use · Independent Assurance Authority · ISA/IEC 62443-3-3 / 4-2 · Product-specific companion to the 62443 assessment tool; confirm exact menu paths / CLI against the deployed firmware and the project hardening baseline. ISOLATED tools run only in a representative/lab environment with written authorization.